

Exact Reliability under Imprecision

Diamonds, intervals, and the p-box dilemma

Temitope Ohiani

University of Strathclyde

July 2026

The problem, and why the obvious thing is wrong

Goal: for every node v , the probability that at least one viable path reaches it from a source:

$$b(v) = P(\text{some source} \rightsquigarrow v).$$

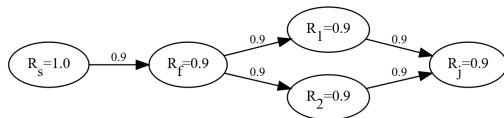
Local message passing:

$b(v) = R(v)(1 - \prod_i (1 - S_i))$... if incoming signals are independent.

Reconvergence breaks it. Both paths share the fork f :

- independent update: $b(j) = 0.749$
- exact (path enumeration): $b(j) = 0.675$

Shared ancestry $\Rightarrow$ positive correlation $\Rightarrow$ the naïve union *overestimates*.



all components 0.9

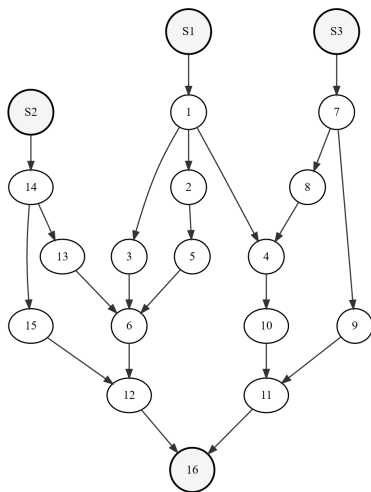
The fix: condition on the fork (a local cutset)

Condition on the shared fork's reachability $\mathcal{R}_f$, then recombine by total probability:

$$b(j) = \underbrace{w}_{P(\mathcal{R}_f=1)} \cdot \psi(1) + (1 - w) \cdot \psi(0), \quad \psi(s) = P(\text{signal reaches } j \mid \mathcal{R}_f = s).$$

- Given the fork's state, the branches are **conditionally independent** $\Rightarrow$ closed-form update applies again (separator sufficiency).
- On the simple diamond: $0.81 \times 0.926559 + 0.19 \times 0 = 0.750513 \Rightarrow b(j) = 0.675$ — exact.
- General case: a separating set C of the parents' shared ancestry, $2^{|C|}$ states.
- This *is* cutset conditioning (Pearl) — specialised: **local** cutsets at each reconvergence, recursively, instead of one global cutset.

Nesting: diamonds inside diamonds



Resolved inner-first; solved diamonds become **supernodes** storing their conditional map $s \mapsto \psi(s)$.

	join	fork	w	$\psi(1)$	$\psi(0)$	$b(j)$
D_1	6	1	0.81	0.859	0.531	0.797
D_2	11	7	0.81	0.804	0.430	0.733
D_3	12	14	0.81	0.839	0.525	0.779
D_4	16	1	0.81	0.823	0.736	0.806

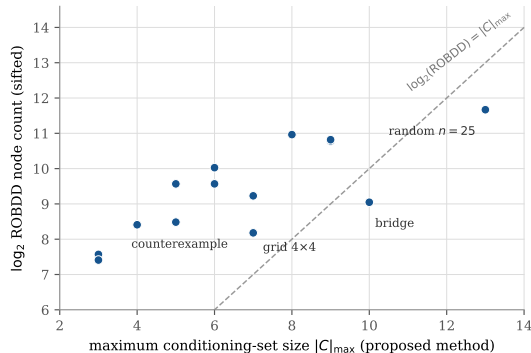
- $\psi(0) \neq 0$: other sources still feed the joins.
- Verified against exhaustive path enumeration: agreement $\leq 3.4 \times 10^{-16}$.
- Reuse matters even here: 68 sub-problems with supernode reuse vs 145 without.

What it costs — and the honest comparison

Per-instance cost is *definite*, known before enumerating:

$$W = \sum_d 2^{|C_d|} \cdot O(|E_d|).$$

- $|C|_{\max}$ tracks the same structural **width** that governs junction trees ($2^{\text{treewidth}}$) and sifted ROBDDs ($2^{\text{pathwidth}}$).
- Measured across the corpus: **neither method dominates**.
- So: no asymptotic advantage for exact point values.
The value has to come from somewhere else.



One recursion, three number types

Everything above is arithmetic over $\{+, \times, 1 - (\cdot)\}$ on probabilities. So swap the number type underneath the *same* recursion:

type	meaning	status
point value p	reliability known	exact (vs ROBDD: 10^{-16} , 129 graphs + families)
interval $[\underline{p}, \bar{p}]$	only bounds known	exact — and the easy case
p-box $(\underline{F}, \bar{F})$	partial distribution known	the hard case — rest of this talk

Epistemic uncertainty (sparse data, expert bounds) rides *through* the analysis instead of being collapsed into a point estimate.

Intervals: the easy case (deliberately anticlimactic)

$b(v)$ is **monotone non-decreasing in every component reliability** (more reliable parts never hurt connectivity). Therefore:

$$[\underline{b}(v), \overline{b}(v)] = [b(\text{all lower corners}), b(\text{all upper corners})] \quad \text{— exactly.}$$

- No dependency dilemma *at all*: the extreme corners realise the extreme outputs.
- Validated against decision-diagram corner evaluation: $\sim 10^{-16}$, whole corpus.
- Ignoring reconvergent dependence instead: over-wide by up to 0.45.
- Cost $\approx 1.2\times$ point-valued ($\approx$ free); one-shot queries $2.9\times$ – $95\times$ faster than build-diagram-then-evaluate-corners (one-shot scope only).

Interval propagation inherits exactness for free. P-boxes will not be so kind.

Where p-boxes break the recombination

The same conditioning step,

$$b = W \cdot A + (1 - W) \cdot B,$$

but now W, A, B are *distributions* (p-boxes) — and A, B are **dependent**: both are functions of the same shared upstream reliabilities.

- This is a **convex combination of dependent random quantities** — not an independent sum.
- Computing it as an independent convolution is **unsound**: measured mass leak up to **0.34** outside $[0, 1]$ on the benchmark grid (the bug we found and fixed).
- A wrong-but-plausible p-box is worse than no p-box: it *certifies* falsehoods.

The correct operator

Integrate over the weight's own distribution, blending the branch distributions at each weight level, enveloped over W 's imprecision:

$$M = \text{env}_W \int \text{blend}(w \cdot A, (1-w) \cdot B) dF_W(w),$$

where **blend** bounds the A – B dependence. Numerical hardening: result intersected with $[0, 1]$ (sound projection — can only tighten a probability), *guarded*: excursions $> 10^{-6}$ raise an error instead of being clamped, so a real regression cannot hide as rounding.

The open choice is the blend — and it is a genuine dilemma: provability versus tightness.

The dilemma: provable vs tight

operator	dependence assumed	soundness	tightness
cvxF	none (Fréchet–Hoeffding)	proven (classical theorem, Williamson & Downs 1990)	conservative; can go vacuous
cvxP	non-negative only	empirical : 0 violations vs MC, all configs; <i>no theorem</i>	tighter; never vacuous in practice

Why non-negative dependence is physically justified: both branches are **monotone-increasing** in the shared upstream reliabilities $\Rightarrow$ they move together $\Rightarrow$ negative dependence is impossible; full Fréchet wastes that half of the dependence range.

Paper stance: **lead with cvxF for every guaranteed claim**; present cvxP as validated, proof open.

The open problem (please attack this)

Claim we cannot yet prove: the positive-dependence blend is sound whenever the branch dependence is non-negative but otherwise unspecified.

- The dependency-bound literature is a strict trichotomy: independence / fully specified copula / fully unknown. **Nothing** in between — checked, not just un-found.
- Natural attempt: supermodular (Tchen) ordering — **obstructed**: the comparison function is not uniformly ordered over outcomes.
- Concrete obstruction: independent vs comonotone uniforms — the combined distributions *cross*; neither uniformly bounds the other.

Until resolved: cvxP stays labelled “empirically validated, proof future work” — never “guaranteed”. A proof (or a counterexample!) is publishable either way.

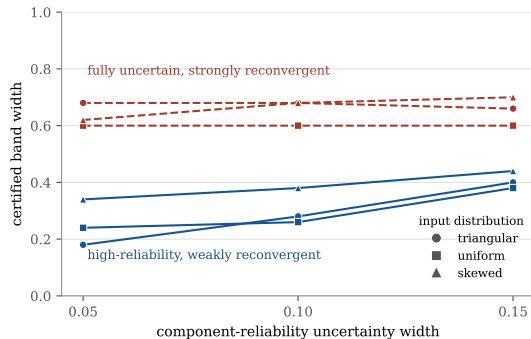
Sound is not tight — the honest envelope

Soundness **never** varies (0 violations, all 18 sweep configs). Tightness does:

- high-reliability, weakly reconvergent: band 0.18–0.44 ($\approx \pm 9\text{--}22\%$) — useful;
- fully uncertain, strongly reconvergent: band 0.60–0.70 ($\approx \pm 30\%$) — conservative.

Driver: reconvergence depth $\times$ input uncertainty. Input *shape* is secondary.

Band is the *worst* threshold's gap — the curves pinch at the ends, where real requirements live.

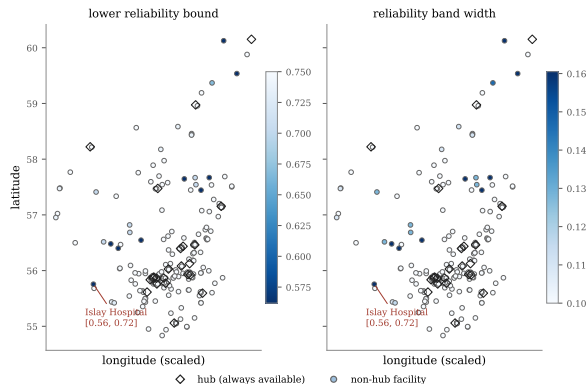


The payoff: a certified decision bound from one propagation

Requirement $x^* = 0.95$ at the benchmark target node:

- certified band on $P(b \leq x^*)$: width **0.02–0.12** — even in the regime whose overall envelope is 0.68;
- Monte Carlo to match: **267–9,604 samples** (worst-case planning), and still only an estimate;
- decision diagrams: **no analytic route at all.**

Applied: Scotland drone network — worst-served facility (Islay Hospital) certified at $[0.56, 0.72]$ reachability.



Limitations, honestly — and takeaways

Limitations

- p-box cost is quadratic in the discretisation level (2.7 s @ 50 $\rightarrow$ 110 s @ 800 on $n=15$): a precision/cost dial, small networks for now.
- Imprecision is *not* a scalability fix: same conditioning depth as exact propagation.
- cvxP soundness proof open (previous slide).

Takeaways

- 1 One exact recursion; conditioning on local cutsets handles reconvergence; same width class as BDDs/junction trees — no free lunch on exact point values.
- 2 Intervals: exactness for free, by monotonicity.
- 3 P-boxes: dependence at the recombination is the whole difficulty; Fréchet blend gives *guaranteed* distributional bounds no exact competitor produces — with a well-characterised tightness envelope and one clean open problem.

Questions — or better, attacks on the open problem.

$W \cdot A + (1 - W) \cdot B$, A, B positively but unspecifiedly dependent:
is the $\text{env}(\text{independent}, \text{comonotone})$ blend always sound?